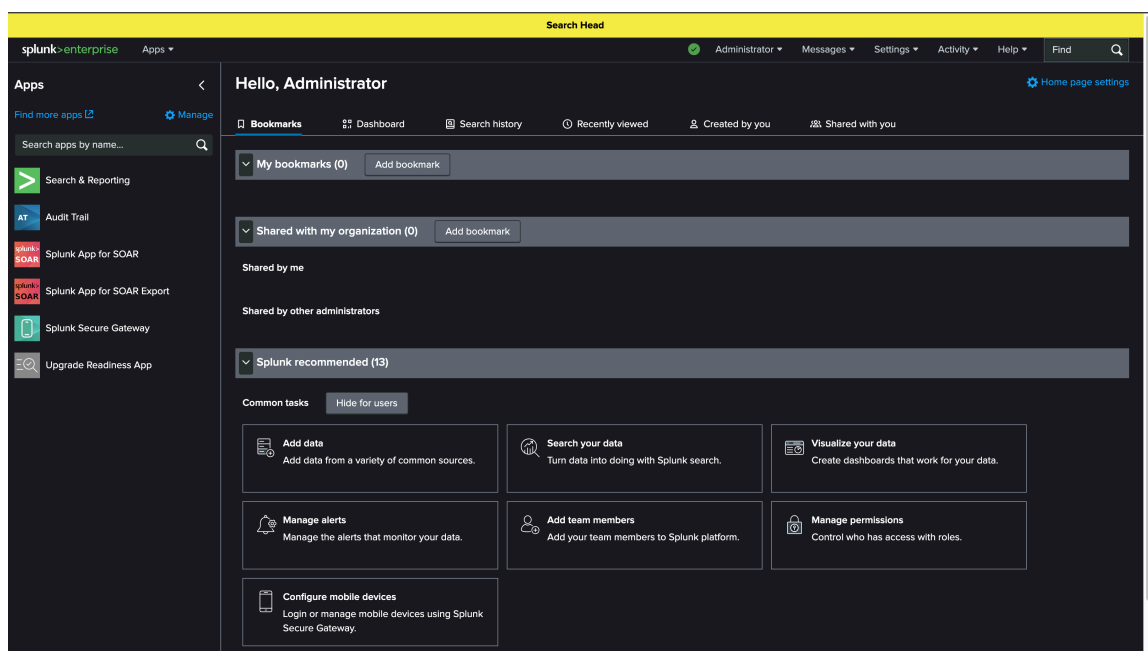


Creating alert for Privilege Escalation Usecase in Splunk Enterprise (Search Head)

Summary

This document explains the steps to create a new Privilege Escalation alert in Splunk Enterprise (Search Head), ensures that the Correlation SPL returns valid results which is later used in Search field of alert configuration and other Configurations needed in alert before sending it to SOAR via Event Forwarding of Splunk App for SOAR Export app.

1. Login as admin in Search Head UI



2. Run this correlation SPL in Search and Reporting app

In Search Head UI,
Go to **Search and Reporting app**,
Run:

```
(index=win_idx EventCode=4732) OR (index=win_ad_idx
EventCode=4728)
| rex "(?is)Member:\s+Security ID:\s+(?<member_sid>S-[0-9\-\
]+)"
| rex "(?is)Group:.*?Group Name:\s+(?<group_name>[^\r\n]+)"
| rex "(?is)Member:.*?Account
Name:\s+cn=(?<added_user1>[^\s,]+)"
| rex "(?is)Member:.*?Account
Name:\s+(?<added_user2>[^\r\n]+)"
| rex "(?is)Subject:.*?Account Name:\s+(?<actor>[^\r\n]+)"
| eval added_user=coalesce(added_user1, added_user2)
| eval added_user=if(added_user=="-" OR added_user=="",
null(), added_user)
| eval user=lower(added_user)
| eval system=coalesce(ComputerName, host)
| eval source=case(EventCode==4728,"DC",
EventCode==4732,"Client")
| eval alert_title="Privilege Escalation Detected"
| table _time alert_title member_sid group_name user actor
system source EventCode
```

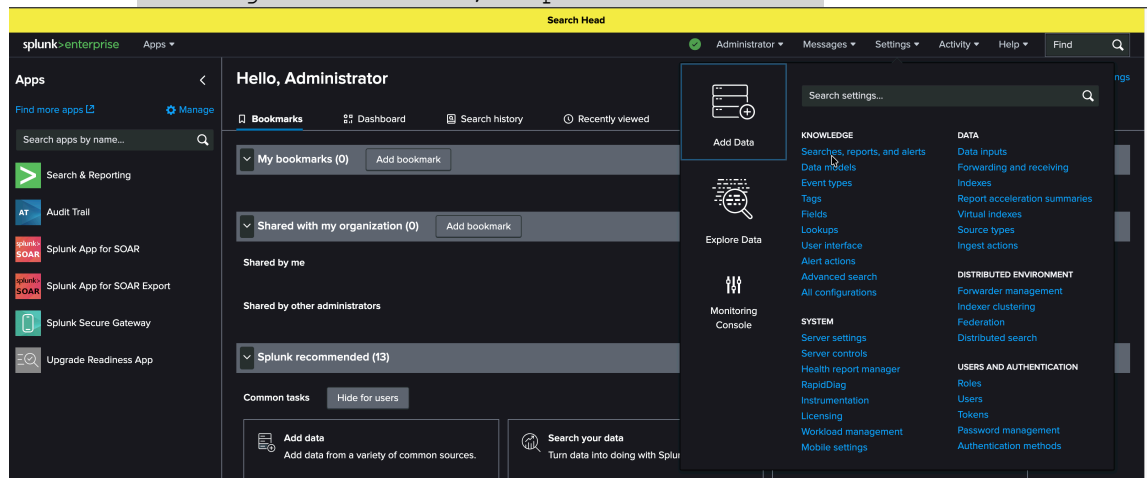
Expected: You should see the logs pointing to the Privilege Escalation incident for both Server and Client.

The screenshot shows the Splunk Search and Reporting app interface. The 'New Search' panel contains the SPL provided in the previous block. Below the search panel, the results are displayed in a table view. The table has columns for _time, alert_title, member_sid, group_name, user, actor, system, source, and EventCode. Two events are shown, both with the alert title 'Privilege Escalation Detected'.

_time	alert_title	member_sid	group_name	user	actor	system	source	EventCode
2026-05-14 14:43:00	Privilege Escalation Detected	S-1-5-21-3821654831-3958487264-1625118876-1125	Administrators	admin	admin	EC2AMAZ-CLIENT01.splunkadmin.local	Client	4732
2026-05-14 14:42:00	Privilege Escalation Detected	S-1-5-21-3821654831-3958487264-1625118876-1125	Domain Admins	eric.wan	admin	EC2AMAZ-DC01.splunkadmin.local	DC	4728

3. Create a new Privilege Escalation alert

- 1) Open **Search Head UI** in a new tab,
Go to **Settings > Searches, reports and alerts**



- 2) Click **New Alert**
- 3) Enter the following in **Create Alert popup**
 - i) In **Alert Settings**,

Alert Title: Privilege Escalation - Domain + Local Admin Detection

Description: Detects users added to Domain Admins and Administrators within same SID correlation window

Search:

```
(index=win_idx EventCode=4732) OR (index=win_ad_idx
EventCode=4728)
| rex "(?is)Member:\s+Security ID:\s+(?<member_sid>S-[0-9\-\]+)"
| rex "(?is)Group:.*?Group
Name:\s+(?<group_name>[^\r\n]+)"
| rex "(?is)Member:.*?Account
Name:\s+cn=(?<added_user1>[^\r\n]+)"
| rex "(?is)Member:.*?Account
Name:\s+(?<added_user2>[^\r\n]+)"
| rex "(?is)Subject:.*?Account
Name:\s+(?<actor>[^\r\n]+)"
| eval added_user=coalesce(added_user1, added_user2)
| eval added_user=if(added_user=="-" OR added_user=="",
null(), added_user)
| eval user=lower(added_user)
| eval system=coalesce(ComputerName, host)
| eval source=case(EventCode==4728,"DC",
EventCode==4732,"Client")
| eval alert_title="Privilege Escalation Detected"
| table _time alert_title member_sid group_name user
actor system source EventCode
```

App: Splunk App for SOAR Export (phantom)

Alert type: Scheduled

Run on Cron Schedule

Time Range: Last 30 minutes

Cron Expression: */10 * * * *

Expires: 24 hours

Create Alert



Settings

Title Privilege Escalation - Domain + Local Admin Detection

Description Detects users added to Domain Admins and Administrators within same SID correlation window

Search

```
(index=win_idx EventCode=4732) OR (index=win_ad_idx EventCode=4728)
| rex "(?is)Member:\s+Security ID:\s+(?<member_sid>S-[0-9\~-]+)"
| rex "(?is)Group:.*?Group Name:\s+(?<group_name>[^\r\n]+)"
| rex "(?is)Member:.*?Account Name:\s+cn=(?<added_user1>[^\,]+)"
| rex "(?is)Member:.*?Account Name:\s+(?<added_user2>[^\r\n]+)"
| rex "(?is)Subject:.*?Account Name:\s+(?<actor>[^\r\n]+)"
| eval added_user=coalesce(added_user1, added_user2)
| eval added_user=if(added_user=="-" OR added_user=="", null(), added_user)
| eval user=lower(added_user)
| eval system=coalesce(ComputerName, host)
| eval source=case(EventCode==4728,"DC", EventCode==4732,"Client")
| eval alert_title="Privilege Escalation Detected"
| table _time alert_title member_sid group_name user actor system source
EventCode
```

App Splunk App for SOAR Export (phantom) ▼

Permissions Private Shared in App

Alert type Scheduled Real-time

Run on Cron Schedule ▼

Time Range Last 30 minutes ▶

Cron Expression */10 * * * *

e.g. 00 18 *** (every day at 6PM). [Learn More](#)

Expires 24 hour(s) ▼

Trigger Conditions

Cancel

Save

ii) **Trigger Conditions**

Trigger alert when: Number of Results > 0

Trigger: For each result

Throttle: On

Suppress results containing field value: member_sid

Suppress triggering for: 10 minutes

iii) **Trigger Actions**

Click +Add Actions > Add to Triggered Alerts

When triggered: Add to Triggered Alerts

Severity: High > Save

Create Alert

×

App	Splunk App for SOAR Export (phantom) ▼	
Permissions	Private	Shared in App
Alert type	Scheduled	Real-time
	Run on Cron Schedule ▼	
Time Range	Last 30 minutes ▶	
Cron Expression	*/10 * * * *	
	e.g. 00 18 * * * (every day at 6PM). Learn More	
Expires	24	hour(s) ▼

Trigger Conditions

Trigger alert when	Number of Results ▼	
	is greater than ▼	0
Trigger	Once	For each result
Throttle ?	☒	
Suppress results containing field value	member_sid	
Suppress triggering for	10	minute(s) ▼

Trigger Actions

+ Add Actions ▼

When triggered	▼ Add to Triggered Alerts Remove
	Severity High ▼

Cancel

Save

4. Ensure proper Permissions are given

After alert creation,

Click **Edit** > **Edit Permissions** of the newly created alert in the table

Check the following:

Display For: All apps

Everyone: Read

admin: Write

Edit Permissions

Alert

Privilege Escalation - Domain + Local Admin Detection

Owner

admin

App

phantom

Display For

Owner

App

All apps

	Read	Write
Everyone	☒	☐
admin	☐	☒
can_delete	☐	☐
phantom	☐	☐
phantomdelete	☐	☐
phantomsearch	☐	☐
power	☐	☐
sc_admin	☐	☐
splunk-system-role	☐	☐
splunk_app_soar_dashboards	☐	☐
splunk_system_upgrader	☐	☐
user	☐	☐

Cancel

Save

Next Steps

Creating an Event Forwarding for sending the key field of the alert SPL to SOAR

Playbook artifact CEF for execution – Refer

26_Creating_EventForwarding_Splunk_App_for_SOAR_Export_app_on_Splunk_Enterprise_V1.pdf.